

BOWEN WU

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Education

ETH Zürich

2023 – 2027 (Expected)

PhD Candidate in Computer Science

Zürich, Switzerland

- Supervised by Prof. Gustavo Alonso and Prof. Ana Klimovic.
- Build databases for heterogeneous computing platforms, including CPUs, GPUs, FPGAs, and advanced networking technologies (e.g., Infiniband, RoCE).
- Explore the intersection of GPU-accelerated data management systems and AI workloads.

ETH Zürich

2020 – 2023

MSc in Computer Science with distinction (GPA: 5.78/6)

Zürich, Switzerland

The Chinese University of Hong Kong

2016 – 2020

BSc in Computer Science, First Class Honor (major-GPA: 3.85/4)

Hong Kong, China

Work Experiences

Meta Platforms, Inc

May 2025 – Aug 2025

Software Engineer Intern

Menlo Park, USA

- Worked with Velox, a high-performance, open-source query execution engine.
- Prototyped a GPU-based reader for the Nimble file format, a replacement for Apache Parquet and ORC. The reader supports important features, such as decoding nested-encoded data, filter push-down, and data chunking.

Microsoft Gray Systems Lab

May 2024 – Aug 2024

Research Intern

Redmond, USA

- Researched and optimized distributed multi-GPU SQL query acceleration.
- Profiled and modeled heterogeneous interconnects (e.g., NVLink, PCIe, Ethernet) in GPU clusters to assess their impact on GPU-accelerated query execution.
- Proposed a novel approach to optimize data exchange over heterogeneous interconnects, significantly enhancing query execution efficiency in GPU clusters. A patent for this technique has been applied for.

Cablex AG (Swisscom)

Aug 2022 – Oct 2022

Database Analyst (Part-time)

Zürich, Switzerland

- Designed a unified relational data model to enable comprehensive data analysis across multiple business sections.
- Leveraged cloud technologies to digitize and modernize operational workflows.

Amazon Web Service

Sep 2021 – Feb 2022

Software Engineer Intern

Berlin, Germany

- Optimized the query rewriter for Redshift, petabyte-scale cloud data warehouse in AWS.
- Proposed a collection of rewriting algorithms for nested SQL queries and implemented them in Redshift using both C++ and a proprietary domain-specific language.
- Improved the performance significantly with my proposed algorithms on the tera-byte scale TPC-DS benchmark.

Publications

- **Bowen Wu**, Marko Kabić, Sven Hepkema, Vasilis Mageirakos, Gustavo Alonso “Eiger: An Efficient Library for GPU-based Data Analytics”, under submission.
- **Bowen Wu***, Wei Cui*, Carlo Curino, Matteo Interlandi, Rathijit Sen “Terabyte-Scale Analytics in the Blink of an Eye”, <https://arxiv.org/abs/2506.09226>, to appear in VLDB 2026.
- Marko Kabić, **Bowen Wu**, Jonas Dann, Gustavo Alonso “Powerful GPUs or Fast Interconnects: Analyzing Relational Workloads on Modern GPUs”, VLDB 2025.
- Vasilis Mageirakos, **Bowen Wu**, Gustavo Alonso “Cracking Vector Search Indexes”, VLDB 2025.
- **Bowen Wu**, Dimitrios Koutsoukos, Gustavo Alonso, “Efficiently Processing Joins and Grouped Aggregations on GPUs,”, SIGMOD 2025.
- **Bowen Wu***, Chenyu Jiang*, Sanghamitra Dutta, Pulkit Grover, “An Information-Theoretic Measure for Enabling Category Exemptions with an Application to Filter Bubbles,” BIAS@ECIR2021.
- Hongzhi Chen, **Bowen Wu**, Shiyuan Deng, et al., “High Performance Distributed OLAP over Property Graphs with Grasper,” SIGMOD 2020 demo track.

Invited Talks

• Velox Tech Talk at Meta	9/26/2025
• Huawei Edinburgh Research Center	3/18/2025
• IBM Research Center	3/3/2025
• Microsoft Gray Systems Lab	6/4/2024

Selected Past Projects

Barrelfish Operating System	Feb 2022 – Jun 2022
<i>Advanced Operating System group project</i>	<i>ETH Zürich</i>
• Learned in-depth about the vast operating system design space, such as micro-kernel, capabilities, and scheduling. • Implemented core functions of Barrelfish operating system for armv8 architecture on QEMU and i.MX 8x SoC, which include memory management and paging, multi-threading, process spawning, multi-core support, intra-/inter-core communication, and a FAT-32 file system. Contributed ~6000 lines of C code individually. • Redesigned the intra-/inter-core communication to be non-blocking with queues, multi-threading, and (de-)multiplexing.	

Teaching

• Data Management Systems - (Head) Teaching Assistant & Invited Speaker	Autumn 2024, 2025
• Hardware Acceleration for Data Processing Seminar - Invited Speaker	Autumn 2024
• Data Modeling and Databases - Teaching Assistant	Spring 2024, 2025, 2026
• Information Systems for Engineers - Teaching Assistant	Autumn 2023

Technical Skills

Languages:	C/C++, CUDA, Python, Shell Script
Systems:	PyTorch, PostgreSQL, SQL Server
Frameworks:	MPI, NCCL, CUB, CuCollection, libcudf, Torch
Tools:	NSight System, NSight Compute
Topics:	GPU Programming, Databases, Distributed Systems, Cloud Computing, Systems for ML



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Department of Computer Science
Institute of Computing Platforms

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Zurich, 8 April 2026

Support Letter for Mr. Bowen WU

Mr. Bowen Wu is currently pursuing a doctoral degree in Computer Science under my supervision at the Department of Computer Science, ETH Zürich. He is in the third year of his PhD and has already demonstrated strong research capabilities, with multiple publications in top-tier data management venues, including VLDB and SIGMOD. I therefore strongly recommend him for the VLDB 2027 Shadow Program Committee.

Should you have any questions, do not hesitate to contact me, or my assistant, Natasha Fischer-Vaidya (natasha.vaidya@inf.ethz.ch).

Sincerely,

Prof. Gustavo Alonso
Head of the Institute of Computing Platforms